

HOMEWORK 2, DUE SEPTEMBER 17

(10 points) 1. Prove that if x and y are arbitrary real numbers such that $x < y$, there exists at least one irrational number z satisfying $x < z < y$.

(10 points) 2. Find the supremum and the infimum of the sets

$$S_1 = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

and

$$S_2 = \left\{ \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}.$$

(10 points) 3. If T, S are nonempty sets so that $s \leq t$ for all $s \in S$ and $t \in T$, prove that $\sup S$ and $\inf T$ exist and prove the inequality $\sup S \leq \inf T$.

(10 points) 4. For A and B nonempty bounded subsets of real numbers, let $A+B$ be the set of all sums $a+b$, where $a \in A$ and $b \in B$. Prove that

$$\sup(A+B) = \sup A + \sup B.$$